

Telco Customer Churn Analysis

Data Science Mini Project — exploratory analysis + churn prediction on 7,043 telecom customers.

Dataset: IBM Telco Customer Churn (public version of the Kaggle dataset)

Stack: Python, pandas, seaborn, matplotlib, scikit-learn

Key findings

- 26.5% of customers churn
- Month-to-month contracts: 42.7% churn vs 2.8% on 2-year contracts
- 52.9% of customers churn within their first 6 months
- Fiber optic users churn at 41.9% (DSL: 19.0%)
- Electronic check payers churn at 45.3%
- No tech support: 41.6% churn vs 15.2% with it

Model

Logistic Regression and Random Forest, both ROC-AUC ≈ 0.84 , recall on churners ≈ 0.74 – 0.78 .

Run it

Open `Telco_Churn_Analysis.ipynb` in Google Colab and run all cells (data loads from a public URL).